

Chapter 2 – Graph Theory

Key Definitions

Network Science · Barabasi et al.

1. Graphs and Networks

Network (Graph) – A catalog of a system's components and the direct interactions between them. Formally, a graph $G = (V, E)$ consists of a set of **nodes** (vertices) V and a set of **links** (edges) E .

Node (Vertex) – A fundamental unit of a network, representing a component of the system (e.g. a webpage, a protein, a person, a router).

Link (Edge) – A connection between two nodes, representing a direct interaction or relationship between the corresponding components.

N – Number of Nodes – The total count of components in the network, also called the **size** of the network. Nodes are labelled $i = 1, 2, \dots, N$.

L – Number of Links – The total count of interactions in the network. A specific link is identified by the pair of nodes it connects, e.g. link $(2, 4)$ connects nodes 2 and 4.

Terminology note: The terms {network, node, link} refer to *real systems*. The terms {graph, vertex, edge} refer to the *mathematical representation* of those systems. In practice they are used interchangeably.

2. Types of Networks

Undirected Network – A network in which all links have no defined orientation. The relationship is symmetric: if node i connects to node j , then j connects to i .
Examples: Internet, power grid, science collaboration networks.

Directed Network (Digraph) – A network in which all links have a specified direction, pointing *from* one node *to* another. The relationship need not be symmetric.
Examples: WWW, mobile phone calls, citation network.

Weighted Network – A network in which each link (i, j) carries a numerical value w_{ij} representing the strength, intensity, or capacity of the interaction.
Examples: Mobile phone calls (minutes), power grid (current flow).

Unweighted (Binary) Network – A network in which links are simply present or absent, carrying no numerical weight. The adjacency matrix contains only 0s and 1s.

Complete Graph (Clique) – A network in which *every* pair of distinct nodes is connected by a link, containing the maximum possible number of links. Average degree: $\langle k \rangle = N - 1$.
Example: All actors sharing a film are mutually connected in the actor network.

Empty Network – A graph with N nodes and **no links** ($L = 0$). Every node is isolated. It is the complement of a complete graph.

Regular Network – A network in which every node has exactly the same degree k , called a k -regular graph. A complete graph K_N is $(N - 1)$ -regular.
Example: A ring lattice where each node connects to its k nearest neighbours.

Cycle Network – A network arranged as a single closed ring where each node connects to exactly two neighbours, forming one large cycle. A cycle network on N nodes is 2-regular with exactly N links.

Sparse Network – A network where the actual number of links L is much smaller than the maximum possible $L_{\max}$. Most real-world networks are sparse.

Multigraph – A network in which multiple parallel links are permitted between the same pair of nodes. Networks that do not allow this are called **simple graphs**.
Example: A social network distinguishing friendship, family, and professional ties simultaneously.

Self-loop – A link that connects a node to itself. Most networks disallow self-loops, but some (e.g. the WWW) permit them.

Complement of a Network ($\bar{G}$) – Given a graph G , its complement $\bar{G}$ has the same node set but contains exactly the links **absent** in G and omits all links **present** in G . Together, G and $\bar{G}$ form a complete graph.

Planar Network – A network that can be drawn on a flat surface such that **no two links cross each other**. Not all networks are planar.
Example: Road maps and many circuit board layouts are planar.

3. Adjacency, Incidence and Subgraphs

Adjacency (Undirected) – Two nodes i and j are **adjacent** if a link directly connects them. In an undirected network adjacency is symmetric.

Adjacency (Directed) – In a directed network, node j is adjacent *to* node i if there is a directed link from j to i . Adjacency is generally asymmetric.

Adjacency Matrix (A_{ij}) – An $N \times N$ matrix encoding the link structure of a network:

- $A_{ij} = 1$ if a link points from node j to node i ; $A_{ij} = 0$ otherwise.
- *Undirected:* $A_{ij} = A_{ji}$ (symmetric). *Weighted:* $A_{ij} = w_{ij}$.

Incidence (Undirected) – A node i and a link e are **incident** if i is one of the two endpoints of e . Two links are incident if they share a common endpoint.

Incidence (Directed) – Node i is incident *from* link e if e points towards i (head), and incident *to* link e if e points away from i (tail).

Incidence Matrix (B_{ie}) – An $N \times L$ matrix of node–link relationships. For undirected networks: $B_{ie} = 1$ if node i is an endpoint of link e , else 0. For directed networks: $B_{ie} = +1$ if i is the head, $B_{ie} = -1$ if i is the tail, else 0.

Subgraph – A graph H is a **subgraph** of G if every node and every link of H also belongs to G . Informally, any portion of a network obtained by selecting a subset of its nodes and/or links.

Induced Subgraph – A subgraph obtained from G by selecting a subset S of nodes and retaining *all* links of G whose both endpoints lie in S . The link set is fully determined by the chosen node set.

Isomorphism – Two graphs G and H are **isomorphic** if there exists a bijection between their node sets that preserves all link relationships – i.e. two nodes are connected in G if and only if their mapped counterparts are connected in H . Isomorphic graphs are structurally identical, differing only in node labelling.

4. Cliques and Graph Colouring

Clique – A subset of nodes in which *every* pair is directly connected – a complete subgraph. A **maximal clique** cannot be extended by adding any further node. The largest clique is the **maximum clique**.

Example: All actors sharing a film form a clique in the actor network.

Network (Graph) Colouring – An assignment of labels (“colours”) to nodes (or links) such that no two adjacent nodes (or incident links) share the same colour. The minimum number of colours needed to validly colour a graph’s nodes is its **chromatic number** $\chi(G)$.

Key result: A graph is bipartite if and only if $\chi(G) \leq 2$.

5. Degree and Degree Distribution

Degree (k_i) – The number of links directly attached to node i – how many other nodes it is directly connected to.

Example: In a call graph, the degree is the number of distinct contacts a subscriber has called.

Average Degree ($\langle k \rangle$) – The mean number of links per node across the entire network. For an undirected network: $\langle k \rangle = 2L/N$.

In-degree (k_i^{in}) – In a directed network, the number of links pointing **towards** node i .
Example: On the WWW, the count of pages that link to a given page.

Out-degree (k_i^{out}) – In a directed network, the number of links pointing **away from** node i .
Example: On the WWW, the number of hyperlinks a page contains pointing to other pages.

Degree Distribution (p_k) – The probability that a randomly selected node has exactly k links. It is the normalised histogram of node degrees: $p_k = N_k/N$, where N_k is the count of degree- k nodes. Must satisfy $\sum_{k=0}^{\infty} p_k = 1$.

Hub – A node whose degree far exceeds the network average. Hubs are a defining structural feature of scale-free networks; their removal can critically disrupt the network.

6. Walks, Paths and Distances

Walk – An ordered sequence of nodes $v_0, v_1, \dots, v_n$ where each consecutive pair (v_i, v_{i+1}) is connected by a link. **Nodes and links may be revisited** any number of times. Length equals the number of links traversed (counting repetitions).

Path – A walk in which **no node (and therefore no link) is visited more than once**. An ordered sequence of distinct nodes where each consecutive pair shares a link. Length equals the number of links.

Cycle – A closed path: starts and ends at the **same node**, with all intermediate nodes distinct and no link repeated.

Eulerian Path – A walk that traverses each **link** exactly once. Exists in an undirected graph if and only if it is connected and has exactly zero or two nodes of odd degree.

Hamiltonian Path – A path that visits each **node** exactly once (but need not traverse every link).

Shortest Path (Geodesic Path, d_{ij}) – The path between nodes i and j with the fewest links – also called the **distance** between them. Multiple shortest paths of equal length may exist. A shortest path never contains loops.

Network Diameter (d_{max}) – The longest shortest path in the network – the greatest distance recorded between any pair of nodes.

Average Path Length ($\langle d \rangle$) – The mean of all pairwise shortest path lengths, computed only over pairs within the same connected component. Captures the typical separation between two randomly chosen nodes.

Breadth-First Search (BFS) – An algorithm for computing shortest paths. From a source node (labelled “0”), it labels direct neighbours as distance 1, then their unlabelled neighbours as 2, and so on. A node’s label equals its distance from the source. Complexity: $O(N + L)$.

7. Connectedness and Connectivity

Connected Network – A network in which a path exists between every pair of nodes. If any pair lacks a connecting path ($d_{ij} = \infty$), the network is **disconnected**.

Connected Component (Cluster) – A maximal subset of nodes such that every pair within the subset is path-connected, but no further node outside the subset can be added while preserving this property.

Giant Component – The largest connected component in a network, typically containing a large fraction of all nodes. All remaining components are comparatively tiny.

Strongly Connected Component (SCC) – In a *directed* network, a maximal subset of nodes such that for **every ordered pair** (i, j) within the subset, there exists a directed path from i to j *and* from j to i . SCCs partition the entire node set of any directed graph. *Note: In undirected networks, the analogous concept is simply a connected component.*

Connectivity – A measure of how robustly a network remains connected when nodes or links are removed.

- **Node connectivity:** The minimum number of nodes whose removal disconnects the network.
- **Link (edge) connectivity:** The minimum number of links whose removal disconnects the network.

Higher connectivity indicates greater fault tolerance.

Bridge – A single link whose removal disconnects the network or increases the number of connected components. A bridge has link connectivity equal to 1.

Cut Vertex (Articulation Point) – A single node whose removal (along with all its links) disconnects the network or increases the number of connected components. The node-analogue of a bridge.

Cutset – A minimal set of links (or nodes) whose removal disconnects the network into at least two components. The **minimum cutset** is the smallest such set, and its size equals the network’s connectivity (link or node, respectively).

8. Trees and Spanning Trees

Tree – A connected, undirected network with **no cycles**. A tree of N nodes always has exactly $N - 1$ links, and there is a *unique* path between every pair of nodes. Every link in a tree is a bridge.

Forest – A graph with no cycles that is not necessarily connected. Each connected component of a forest is a tree.

Spanning Tree – A subgraph of a connected network G that (i) includes **all** N nodes of G , (ii) is connected and acyclic, and therefore (iii) contains exactly $N - 1$ links. It is the minimal subgraph that reaches every node.

Application: Used in routing protocols and network design to ensure full reachability with minimum redundancy.

9. Clustering Coefficient

Local Clustering Coefficient (C_i) – The fraction of pairs among node i 's k_i neighbours that are actually connected to each other.

- $C_i = 0$: none of node i 's neighbours are connected to each other.
- $C_i = 1$: all neighbours of node i form a complete graph.
- Probabilistic interpretation: C_i is the probability that any two of node i 's neighbours are also directly linked.

Note: Defined as zero for nodes with degree $k_i \leq 1$.

Average Clustering Coefficient ($\langle C \rangle$) – The mean of the local clustering coefficients over all nodes. It is the probability that two neighbours of a randomly selected node are also connected to each other.

Global Clustering Coefficient (C_Δ) – Three times the number of closed triangles divided by the total number of connected triplets in the network. Also called the **ratio of transitive triplets**. Not equivalent to $\langle C \rangle$.

Connected Triplet – An ordered set of three nodes A, B, C where A connects to B and B connects to C (but A and C need not be directly connected). A closed triangle contains three such triplets.

10. Bipartite and k-Partite Networks

Bipartite Graph (Bigraph) – A network whose nodes split into two disjoint sets U and V , where every link connects a U -node to a V -node. No $U-U$ or $V-V$ links exist. Equivalently: a graph is bipartite if and only if it contains **no odd-length cycles** (it is 2-colourable).

Example: Hollywood network – U = movies, V = actors; a link exists if the actor appeared in the film.

Projection (of a Bipartite Network) – A derived network connecting nodes of the same set if they share a common neighbour in the opposite set.

- ***U*-projection:** Two *U*-nodes are linked if they share a *V*-neighbour.
- ***V*-projection:** Two *V*-nodes are linked if they share a *U*-neighbour.

***k*-Partite Network** – A generalisation of bipartite networks to *k* disjoint node sets $V_1, \dots, V_k$, where links only connect nodes belonging to *different* sets. A bipartite graph is the special case $k = 2$.

Example: A tripartite ($k = 3$) recipe-ingredient-flavour-compound network.

Multipartite Network – A *k*-partite network for any $k \geq 2$; the general term for networks with multiple disjoint node classes where links only cross class boundaries.

11. Additional Concepts

Metcalfe’s Law – The value of a communication network is proportional to the square of its number of users ($\sim N^2$), while operating cost grows only linearly ($\sim N$). Based on the observation that possible pairwise connections grow as $N(N - 1)/2$.

Caveat: Violated in practice because real networks are sparse and links have unequal weights.

Small-World Property – In many real networks the average path length $\langle d \rangle$ is surprisingly short (often logarithmic in N), meaning any two nodes are separated by only a small number of intermediate links. Commonly associated with “six degrees of separation.”

Wiring Diagram – An informal term for the complete map of nodes and links of a network, fully describing how all components are interconnected.